

## Complex Engineering Problem

Course Code and Title: EE2001 – Electronic Circuit Design

Semester: Fall 2024

Instructor: Dr. Omer Saleem

Total Marks: 50

Deadline: Tuesday, 10<sup>th</sup> Dec. 2024

### CLOs and PLOs for Complex Engineering Problem

| CLOs  |        | Description                               | Taxonomy Levels | PLO |
|-------|--------|-------------------------------------------|-----------------|-----|
| CLO 2 | Theory | Design electronic circuits using op-amps. | C5              | 3   |

### Instructions

Attempt all parts of the following design problem.

### Problem Statement

Design a biomedical signal conditioning system to process and analyze Electromyogram (EMG) signals using operational amplifiers. EMG signals, typically in the range of 0.1 mV to 5 mV and 10 Hz to 500 Hz, are used to study the bicep muscle activity. The circuit should include multiple stages to amplify and condition the raw EMG signal while enabling feature extraction for diagnostic or control purposes.

- The system uses two EMG sensors. One EMG sensor measures the muscle activity of elbow. It delivers the reference voltage signal  $V_{ref}$ . The other EMG sensor measures the muscle activity of bicep and delivers  $V_{sig}$ . The difference of these signals is instrumentally amplified by a gain of [your batch number] to acquire a differential EMG signal,  $V_D$ .
- The differential EMG signal,  $V_D$ , is passed through a customized circuit that integrates  $V_D$  to compute the total muscle activity over time, and also differentiates it to detect rapid changes in muscle activity. Their combined effect is unified via the following expression.

$$V_E = k_1 \int V_E dt + k_2 \frac{dV_E}{dt} + k_3$$

where,  $k_1$  = first two digits of your 4 - digit roll number

$k_2$  = last two digits of your 4 - digit roll number

$k_3$  = sum of all digits of your 4 - digit roll number

- Finally, the refined muscle activity signals  $V_E$  is **squared** to compute the power of the EMG signal,  $V_O = V_E^2$ , for energy-based analysis.

## Objectives

1. Design an op-amp circuit to implement all the intermediate stages of signal conditional discussed above (starting from Stage (a) to Stage (c)). Your design should include schematic diagrams for each stage with detailed annotations labeling each stage.
2. Evaluate all the resistors, capacitors, and other circuit components used in the circuit. Show complete mathematical calculations to support your choice.
3. State the issues (or imperfections) linked with the circuits in the intermediate stages and suggest plausible remedies.
4. Operate within a limited power budget suitable for portable or wearable medical devices.

## Deliverables

- Schematic diagrams for each stage with detailed annotations and theoretical analysis for the evaluation of circuit components
- Design Report discussing the design strategy, choice of op-amp principles, challenges, and final performance metrics. Your report should include following points:
  - Address the challenge of modular design while maintaining system integration.
  - Provide proper justifications of each op-amp circuit configuration used to fulfill the desired objectives for the circuits in each stage.
  - Comment on the alternative circuit designs for a given stage.
  - Discuss the impact of tolerances of the components on the output voltage

|                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |       |       |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------|-------|
| <b>WP1:</b> Depth of knowledge<br><b>WP2:</b> Range of conflicting requirements<br><b>WP3:</b> Depth of analysis<br><b>WP4:</b> Familiarity of issues<br><b>WP5:</b> Extent of applicable codes<br><b>WP6:</b> Extent of stakeholders<br><b>WP7:</b> Interdependence | Please fill according to the WPs covered in the course CEP, example is shown here. <ul style="list-style-type: none"> <li>• <b>WP1: Depth of Knowledge</b> – Analysis and choice of op-amp principle for each stage requires knowledge of WK3, WK8. Design requires knowledge of WK5, WK8</li> <li>• <b>WP2: Range of conflicting requirements</b> -- Requirements are conflicting. All objectives are desired, but power consumption should be minimized and DC imperfections must be dealt with. The initial amplification stage design is also critical.</li> <li>• <b>WP3: Depth of analysis</b> -- No unique solution. Multiple designs possible. Analysis needed to select the most appropriate one.</li> </ul> |          |       |       |
|                                                                                                                                                                                                                                                                      | Rubrics                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          | CLOs  | Marks |
|                                                                                                                                                                                                                                                                      | Understands objectives, requirements, and constraints                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | WP1, WP2 | CLO 2 |       |
|                                                                                                                                                                                                                                                                      | Clearly explains the op-amp circuit principles for each stage.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | WP1      | CLO 2 |       |
|                                                                                                                                                                                                                                                                      | Considers multiple solutions and selects suitable one based on analysis                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | WP1, WP3 | CLO 2 |       |
|                                                                                                                                                                                                                                                                      | Accurately calculates the circuit components.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | WP1, WP3 | CLO 2 |       |
|                                                                                                                                                                                                                                                                      | Develops a final design which satisfies all requirements                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | WP1, WP3 | CLO 2 |       |
|                                                                                                                                                                                                                                                                      | Report                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |          |       |       |

### Knowledge Profiles

**WK1 - Natural Sciences:** A systematic theory-based understanding of natural sciences applicable to the discipline.

**WK2 - Mathematics and Computing:** The concept-based mathematical thinking, numerical analysis, statistics and formal aspects of computer and information science to support analysis and modelling applicable to the discipline.

**WK3 - Engineering Fundamentals:** A systematic, theory-based formulation of engineering fundamentals required in an engineering discipline.

**WK4 - Engineering Specialization:** The knowledge of Engineering specialization that provides theoretical frameworks and bodies of knowledge for the accepted practice areas that are at the forefront in a discipline.

**WK5 - Engineering Design:** The *Design Thinking Knowledge* that supports engineering design in a practice area of an engineering discipline.

**WK6 - Engineering Practice:** The Knowledge of engineering practices (technology) in different practice areas of an engineering discipline.

**WK7 - Engineering in Society:** A systematic, comprehension-based knowledge of the role of engineers in a society and the professional issues related to practicing engineering profession in a discipline: ethics and the professional responsibility of an engineer to public safety including the impact of an engineering activity i.e. economic, social, cultural, environmental and sustainability

**WK8 - Research Literature:** Engagement with selected knowledge in the research literature of the discipline.



Name: Sheza Naveed

Roll No.: 22L-7651

Section: 5A

Assignment: 4 (Electronic Circuit Design)

a) Given data:

$V_{ref}$ : EMG signal from elbow sensor

$V_{sig}$ : EMG signal from bicep sensor

Required differential gain: 22 (batch number)

Circuit Design:

A three op-amp instrumentation amplifier is ideal for high-input impedance and excellent CMRR

- Two non-inverting buffers for  $V_{ref}$  and  $V_{sig}$  to ensure high impedance.
- A differential amplifier using the buffered outputs.

Calculations:

$$G = 1 + \frac{2R_2}{R_1}$$

$$G = 22$$

$$22 = 1 + \frac{2R_2}{R_1}$$

$$2R_2 = 21$$

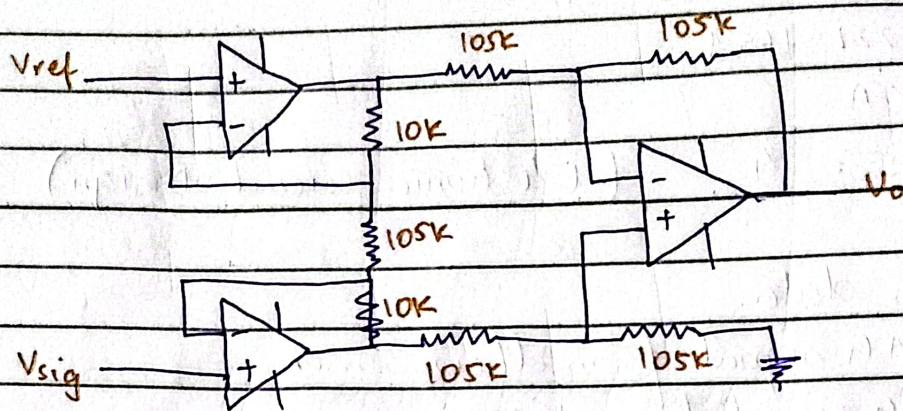
$$R_1$$

$$R_2 = 10.5 R_1$$

Choosing

$$R_1 = 10K\Omega \text{ and } R_2 = 105K\Omega$$





b) This system combines integration and differentiation stages to evaluate total muscle activity and rapid changes.

$$V_E' = k_1 \int V_E dt + k_2 \frac{dV_E}{dt} + k_3$$

From the roll number 7651:

$k_1 = 76$  (First two digits)

$k_2 = 51$  (last two digits)

$k_3 = 19$  (sum of digits)

Integration stage:

An integrator circuit with gain  $k_1 = 76$

$$V_{int} = \frac{1}{R_c} \int V_0 dt$$

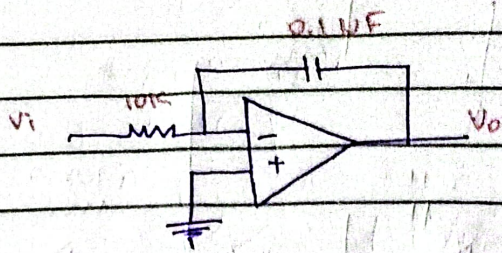
choosing  $R_{int} = 10k \Omega$

$$k_1 = 1$$

$$R_c$$

$$C_{int} = 0.1 \mu F$$





Differentiation stage:

A differentiator circuit with gain  $K_2 = 51$

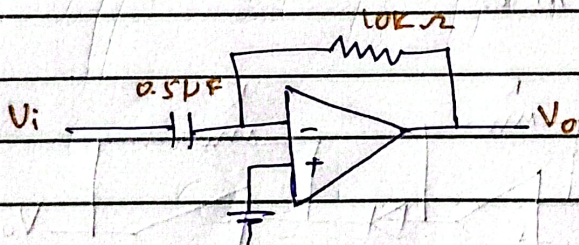
$$V_{diff} = -RC \frac{dV_o}{dt}$$

choosing  $R_{diff} = 10K\Omega$

$$K_2 = R_{diff} C_{diff}$$

$$51 = 10K\Omega C_{diff}$$

$$C_{diff} = 0.5\mu F$$

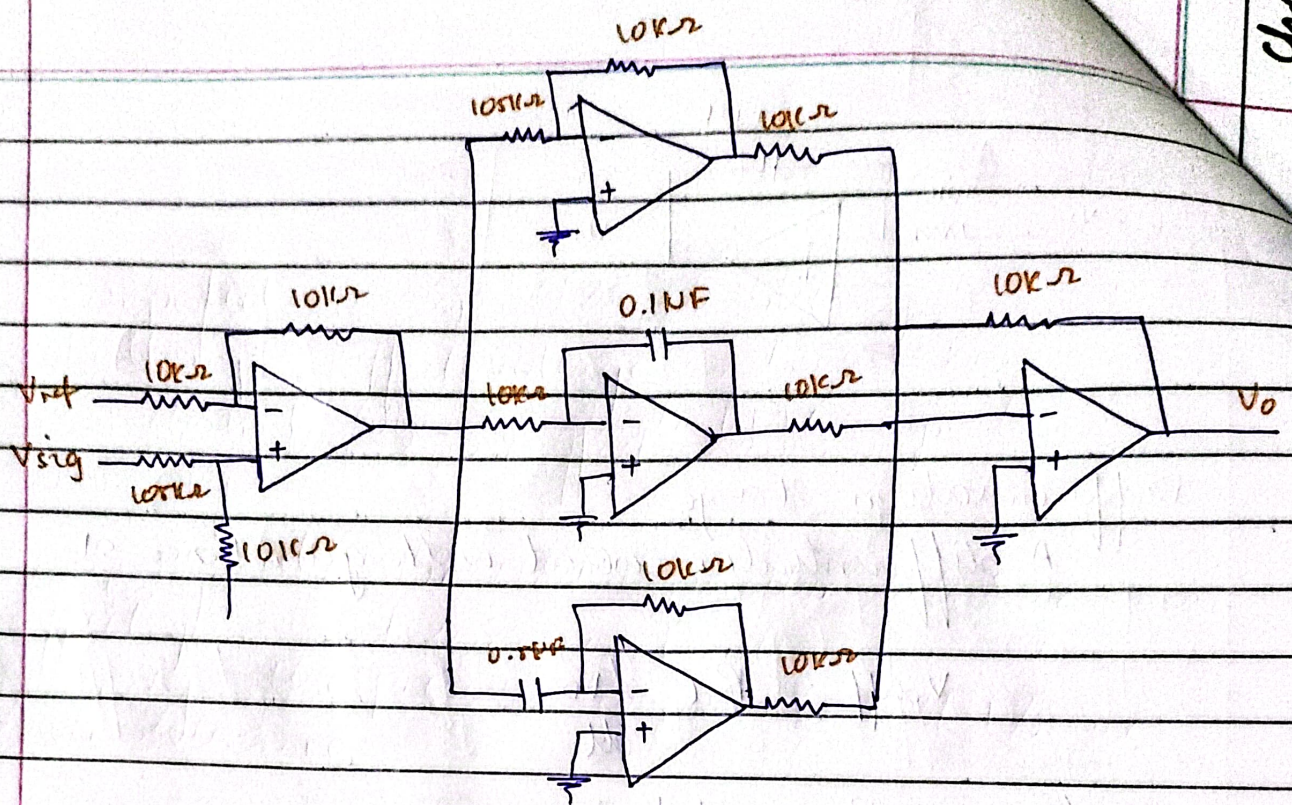


Combination stage:

The combination stage sums up the outputs of the integrator and differentiator using a summing amplifier.

- Use  $R_{sum} = 10K\Omega$  for equal weighting of components

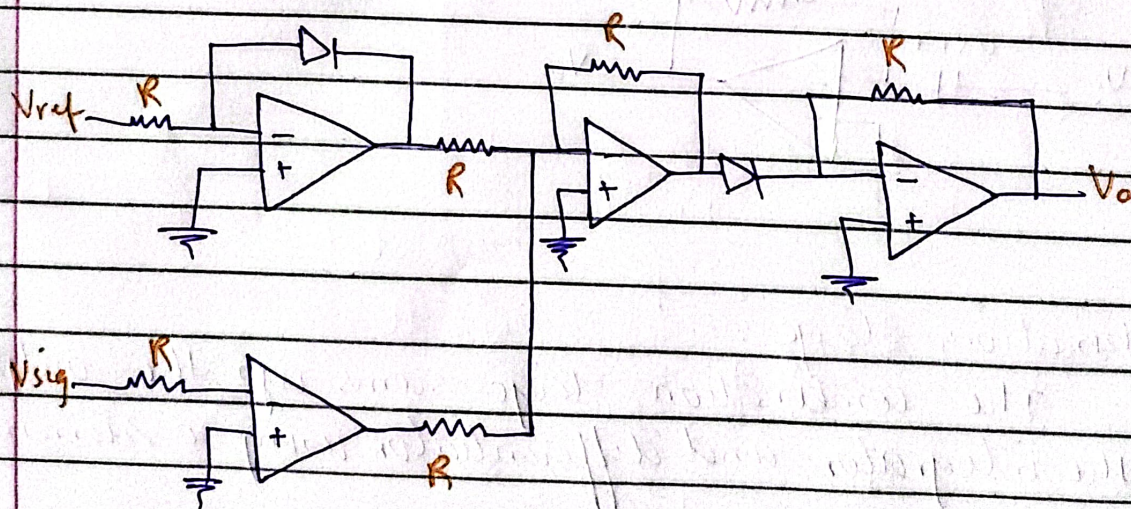




c) To compute the EMG signal power

$$V_{0E} = V_E^2$$

the circuit diagram is



choosing  $R = 10k\Omega$



## Challenges And Remedies

### 1. Imperfect Op-amp characteristics:

**challenge:** Offset voltage and input bias currents can distort low amplitude EMG signals.

**Remedy:** Use low-offset, low noise op-amps.

### 2. Component Tolerances:

**challenge:** Resistor and capacitor tolerances can affect gain and time constants.

**Remedy:** Use precision components with 1% or lower tolerance.

### 3. Power Consumption:

**challenge:** Excessive power usage is unsuitable for wearable device.

**Remedy:** Use low-power op-amps and optimise bias currents.



Name: Sheza Naveed  
Roll No. : 22L-7651  
Section: 5A

## **Assignment Report**

### **1. Introduction**

Electromyography (EMG) is a vital tool in studying muscle activity, where the analysis and conditioning of EMG signals are critical for accurate diagnostics and control applications. This report outlines the design of a biomedical signal conditioning system to process and analyze EMG signals using operational amplifiers (op-amps). The system has three main stages: differential amplification, signal integration and differentiation, and energy-based analysis.

### **2. Design Strategy**

The design focuses on creating a modular yet integrated system that processes two EMG signals:

$V_{sig}$  from the bicep and  $V_{ref}$  from the elbow. A robust op-amp-based circuit is employed to perform the following tasks:

- Differential amplification to extract the EMG signal of interest.
- Signal integration and differentiation to compute total muscle activity and detect rapid changes.
- Squaring the processed signal for energy-based analysis.

### **3. Circuit Design**

#### **3.1 Stage (a): Differential Amplifier**

The first stage is an instrumentation amplifier with a gain determined by the batch number ( $G = \text{Batch Number}$ ). The circuit amplifies the difference  $V_{sig} - V_{ref}$  while rejecting common-mode signals such as noise.

#### **Schematic Diagram**

- Input:  $V_{sig}$  and  $V_{ref}$ .
- Components: Precision op-amps (e.g., AD623 or TL082), matched resistors for gain accuracy, and low-pass filters for noise reduction.

#### **Mathematical Analysis**

$V_D = G(V_{sig} - V_{ref})$  The gain  $G$  is determined by:

$G = 1 + 2R_1/R_{gain}$  where  $R_{gain}$  is calculated based on the batch number.



### 3.2 Stage (b): Integration and Differentiation

The amplified differential signal  $V_D$  is processed further:

1. **Integrator:** Computes total muscle activity over time.
  - Circuit: Integrator using an op-amp, resistor  $R_i$ , and capacitor  $C_i$ .
  - Transfer function:  $V_{int} = k_1 \int V_D dt$  where  $k_1$  is determined by the first two digits of the roll number.
  - $R_i$  and  $C_i$  are chosen to set the integration time constant.
2. **Differentiator:** Detects rapid changes in muscle activity.
  - Circuit: Differentiator with  $R_d$  and  $C_d$ .
  - Transfer function:  $V_{diff} = k_2 dV_D/dt$  where  $k_2$  is the last two digits of the roll number.
3. **Combined Signal:** A summing amplifier combines the integrated and differentiated signals:  $V = k_1 \int V_D dt + k_2 dV_D/dt + k_3 V_D$

### 3.3 Stage (c): Energy Computation

The refined signal  $V$  is squared to compute the power:  $V_oE = V^2$

- Circuit: Precision squaring circuit using analog multipliers (e.g., AD633).

### 4. Component Selection and Calculations

- **Resistors:** Chosen based on gain and stability requirements.
- **Capacitors:** Selected for desired time constants in integration and differentiation circuits.
- **Op-amps:** Low-noise, high-precision op-amps with sufficient bandwidth to handle the EMG frequency range (10 Hz–500 Hz).

### 5. Challenges and Remedies

1. **Noise Sensitivity:**
  - Issue: EMG signals are prone to noise.
  - Remedy: Use low-pass filters and shielded cables to minimize interference.
2. **Component Tolerances:**
  - Issue: Variations in resistor and capacitor values can affect accuracy.
  - Remedy: Use precision components and account for tolerances in the design.
3. **Power Efficiency:**
  - Issue: Portable devices require low power consumption.



- Remedy: Use low-power op-amps and optimize the supply voltage.

#### 4. **Modular Design:**

- Challenge: Ensuring seamless integration.
- Remedy: Design interstage buffers to maintain signal integrity.

#### 6. **Performance Metrics**

- **Bandwidth:** Covers the EMG range (10 Hz–500 Hz).
- **Gain:** Precisely set for differential amplification.
- **Noise Rejection:** Achieved via instrumentation amplification and filters.
- **Power Consumption:** Suitable for portable applications (<1 mW per stage).

#### 7. **Alternative Circuit Designs**

- **Differential Amplifier:** Replace with a programmable gain amplifier (PGA) for adjustable gain.
- **Integrator and Differentiator:** Use digital signal processing (DSP) for improved flexibility.

#### 8. **Conclusion**

The designed system effectively amplifies, conditions, and analyses EMG signals for muscle activity assessment. By addressing the challenges of noise, power, and integration, the circuit achieves reliable performance suitable for wearable biomedical devices. Further optimisation in component selection and layout design can enhance system robustness and usability.